

Plunder, Indirect Rule, and the Limits of Co-optation:

A Strategic Model of Conquest under First-Strike Advantage and Commitment Failure

Sunny Li*

May 13, 2026

Abstract

Why do conquerors plunder some polities, install indirect rule through co-opted local elites in others, and impose direct foreign administration in still others? We present a theory of the conqueror’s institutional choice that focuses on the strategic interaction between a foreign force, the local elite faction it might co-opt, and the rival faction whose elimination is the price of cooperation. When local elites retain enough residual military capability to deter the conqueror’s post-victory expropriation, the cooperation contract is credible and indirect rule under a co-opted partner is the equilibrium; when local elites are too weak to deter expropriation and no inheritable bureaucracy is available, the conqueror plunders rather than co-opts, and the polity records no installed governance at all. We assess the theory using a hand-coded panel of 122 Mongol conquest events between 1211 and 1302, a setting in which the conquering power faced sharply varying levels of pre-conquest local elite strength and inheritable administrative capacity across cities and across periods. Using comparative statics on city size, region, period, and the location of preexisting bureaucracy, we show that plundered cities have a median sustained Mongol control duration of zero years while indirectly and directly ruled cities have median durations of 108 and 117 years; the plunder-versus-co-optation pattern matches the model on every dimension that the standard binary invade-or-co-opt framework cannot accommodate.

Keywords: conquest, coalition formation, indirect rule, commitment problem, Mongol empire.

JEL: D74, F51, N40, P48.

1. Introduction

What determines whether a conqueror plunders, installs direct rule, or co-opts a local elite under indirect rule? Many scholars have argued that direct rule produces stronger states and better long-run development outcomes than indirect rule (Garfias and Sellars, 2021, 2022; Gerring et al., 2011; Spencer Hartnett and Saleh, 2024), and that the institutional form a conqueror imposes shapes the trajectory of state capacity for centuries thereafter (Padró i Miquel and Yared, 2012; Gailmard, 2024). By contrast, indirect rule through co-opted local elites has been the historical norm across empires from the Roman provinces to the Ottoman millets to the British colonial governorates (Gerring et al., 2011; Garfias and Sellars, 2022), and the analytical tradition that grew up around it treats the conqueror’s binary choice as “invade unilaterally or co-opt” (Padró i Miquel and Yared, 2012; Niou and Zeigler, 2019). Yet 53% of recorded Mongol conquest events

*Department of Political Science, Stanford University. Email: dionysus@stanford.edu. I thank Luis Bosshart for helpful discussion. All errors are my own.

between 1211 and 1302 produced no installed governance at all: the army sacked the city, took the surplus, and moved on, leaving zero years of subsequent sustained political control.¹ Plunder of this kind is not direct rule (no administration is installed) and not indirect rule (no local elite is empowered); it is a third action that the binary framework cannot accommodate, and it accounts for the modal Mongol response. It is thus important to understand when conquerors plunder rather than govern, when they govern through co-opted elites rather than through foreign administrators, and what configuration of local elite strength and pre-existing administrative capacity drives the choice.

In this paper, we present a theory of the conqueror’s institutional choice and provide evidence on the determinants of plunder, indirect rule, and direct rule across 122 Mongol conquest events from 1211 to 1302. Building on past work on indirect rule and dynamic coalition formation (Padró i Miquel and Yared, 2012; Acemoglu et al., 2012; Bueno de Mesquita and Lalman, 1988, 2008; Weingast, 1997), we offer an explanation of conquest mode that highlights the strategic dilemma faced by a conqueror who gains additional decision-making power from victory and therefore cannot credibly commit, before victory, to refrain from using that power against the very local elites whose cooperation made victory possible. Direct rule allows the conqueror to capture an annuity of extraction $(1-\kappa_D)/(1-\delta)$ in perpetuity, but it requires a costly inheritable administrative apparatus and exposes the conqueror to insurgency by displaced elites; indirect rule allows the conqueror to delegate administration at near-zero installation cost, but it requires sharing revenue with a local elite who has every reason to suspect the deal will not survive the elimination of its rival (Garfias and Sellars, 2021; Padró i Miquel and Yared, 2012). Two frictions drive the analysis: a battlefield first-strike advantage λ that can be cashed only in violent action and that cannot be brought to the bargaining table, and a post-victory commitment problem in which the conqueror, having gained the additional decision-making power that comes with eliminating the rival faction, cannot bind itself to honour the tax share τ promised to its cooperating partner (Acemoglu et al., 2012; Weingast, 1997). A configuration in which local elites retain enough residual military capability to deter post-victory expropriation can therefore sustain co-opted indirect rule; a configuration in which they do not, and in which no inheritable bureaucracy makes direct administration cheap, collapses to plunder, even when both the conqueror and the local elite would prefer indirect rule *ex ante*. Our theory thus highlights the role that the post-victory commitment problem—not the mere strength of the target—plays in producing one-shot plunder rather than installed governance, and it identifies a sharp, non-monotone relationship between local elite strength and the conquest mode that is unidentifiable in any model that conflates plunder with direct rule. This relates to ongoing debates on whether conquest mode is best modeled as a static bargaining problem (Padró i Miquel and Yared, 2012; Niou and Zeigler, 2019) or as a dynamic coalition-formation problem with endogenously-shifting political power (Acemoglu et al., 2012; Schweller, 1994; Wolford, 2015).

We provide support for the theory using a hand-coded panel of 122 Mongol conquest events between 1211 and 1302, the period spanning Genghis Khan’s western campaigns through the post-Yuan consolidation under Khubilai. After Genghis Khan unified the steppe in 1206, the Mongol military apparatus was equipped with an extreme first-strike advantage—feigned retreat, encirclement, and rapid cavalry concentration (May, 2016; Barfield, 1989a,b)—but faced target polities that varied widely in the autonomous military capability of their local elites and in the depth of their inheritable administrative apparatus (Biran, 2005; Twitchett et al., 1978; Spencer Hartnett and Saleh, 2024). The Khwarezmian empire in 1219, the Jin successor states in 1234, the Persian plateau under the Ilkhanate after 1256, and the Song administrative apparatus absorbed by the Yuan after

¹Hand-coded from Jaques (2006) and Chandler and Fox (1974), $N = 122$; see Section 6.

1279 each present a different combination of these two parameters and each was met with a different conquest mode. Crucially, the cross-sectional and temporal pattern of plunder versus governance maps onto the model’s parameter regions: cities in the Khwarezmian corridor in 1219–1221, where Sultan Muhammad’s pre-conquest purges had gutted the autonomous military aristocracy, were sacked with zero years of subsequent governance; cities on the Persian plateau, where the Juvayni and Rashid al-Din administrative networks survived intact, were absorbed into Ilkhanid indirect rule for over a century (Biran, 2005; Weatherford, 2005). The Mongol–Song coalition of 1233–1234 against the Jin provides a real-time observation of the commitment friction itself: the Mongols proposed a territorial split, the Song accepted, the Jin fell in February 1234, and within a year the Mongols had moved into Henan in violation of the agreed division. The forty-three-year Mongol–Song war that followed is the model’s incentive-compatibility lower bound failing in the historical record (Twitchett et al., 1978). We empirically demonstrate the link between local elite strength, inheritable bureaucracy, and conquest mode using two strategies. We first test the model’s comparative statics on the Mongol panel, showing that mode of rule sorts on city size, region, period, and pre-existing administrative capacity in the directions the model predicts. To extend the test out of the Mongol sample, we sketch a cross-civilizational specification using the Seshat Global History Databank (Turchin et al., 2015; François et al., 2016), which codes 264–620 polities on roughly 1,500 variables including administrative levels, bureaucracy characteristics, professional military variables, and suprapolity relations. We find that plundered cities have a median sustained Mongol control duration of zero years, while indirectly and directly ruled cities have median durations of 108 and 117 years respectively, and that the plunder-versus-governance share shifts from 63% in the early period (1211–1239) to 41% in the late period (1240–1302) as the first-strike advantage λ depreciated and the discount factor δ rose with the stabilization of Mongol succession. The duration gap—zero years versus a century—is robust to alternative codings of conquest mode and is unexplainable in any model that conflates one-shot extraction with installation of direct rule. The particular pattern of plunder concentration in the Khwarezmian corridor, indirect rule clustering on the Persian plateau, and direct rule clustering in north China cannot be explained by competing theories that read the early-Mongol record as irrationality (Kalyvas, 1999; Spruyt, 1994) or as a transitional inefficiency to be overcome by later institutional learning (Turchin, 2009; May, 2016); the pattern is the equilibrium of a four-action choice in which raid is a distinct strategy with its own comparative statics. We further show that the effect of local elite strength on the conquest mode is non-monotone—very weak elites produce plunder, intermediate elites produce indirect rule, very strong elites deter conquest entirely—and that this non-monotonicity, which the standard balance-of-power literature (Walt, 1985; Waltz, 2001; Morgenthau, 1948) predicts to be monotone, is the cleanest discriminating implication of the commitment-failure mechanism.

This paper contributes to classic debates in political economy about the strategic foundations of imperial expansion and the institutional consequences of conquest. Past scholarship on conquest and indirect rule has emphasized the role of monitoring frictions in a static principal-agent problem in which the conqueror’s commitment to honour the agent’s reward is taken as exogenously given (Padró i Miquel and Yared, 2012); in this framing, indirect rule is the optimal contract under information asymmetry, and the question of whether the conqueror would actually honour the contract *ex post* does not arise. Though much work has found evidence in support of the static principal-agent framing in colonial settings (Padró i Miquel and Yared, 2012; Gerring et al., 2011; Garfias and Sellars, 2021), its applicability is questionable in contexts of rapid imperial expansion—Mongol, Tatar, post-Roman—in which the conqueror’s post-victory military advantage over the cooperating partner is large and the cooperation contract has no third-party enforcer. By contrast, we provide a rationale for conquest mode that focuses on the role of the post-victory commitment

problem, both in the form of elite resistance to expropriation and in the form of the conqueror’s own inability to bind itself, and that derives a third equilibrium action—plunder—that the static framing collapses onto direct rule. The relationship between dynamic political power shifts and the stability of cooperation contracts is central to work on modern as well as historical state formation (Acemoglu et al., 2012; Weingast, 1997; Bueno de Mesquita and Lalman, 2008; Gailmard, 2024). The literature, however, is divided on whether the post-victory commitment problem is best modeled as binding only at the margin (Padró i Miquel and Yared, 2012; Niou and Zeigler, 2019) or as the central friction that determines institutional form (Acemoglu et al., 2012; Weingast, 1997). In this paper, we highlight the role of local elites as strategic intermediaries whose residual military capability is the conqueror’s only credible commitment device, and illustrate how the absence of such a capability—not the presence of a strong target—produces plunder rather than installed governance.

Our focus on the post-victory commitment problem also sets this paper apart from the classic literature on nomadic conquest and the steppe frontier, and particularly work linking nomadic empire formation to the wealth gradient at the agrarian frontier (Barfield, 1989a,b; Turchin, 2009). In these accounts, the institutional impacts of nomadic conquest operate largely through the geography of the steppe-sown interface, the opportunistic timing of raids when sedentary states are weak, and the recurrent collapse of nomadic confederations after the founder’s death. This in turn produces a coarse periodization of nomadic empire as “raid-then-settle,” with the raiding phase treated as preparatory and the settlement phase as the genuine institutional outcome. Plunder serves a different role in our theory. Consistent with the historical evidence on the Khwarezmian campaign and the Mongol–Song war and a growing literature on commitment failure in dynamic coalitions (Acemoglu et al., 2012; Schweller, 1994; Wolford, 2015), we argue that plunder is the equilibrium outcome of a parameter region in which no co-optation contract is incentive-compatible. Like other shocks to the credibility of cooperation contracts, the absence of a residual deterrent makes the conqueror’s promise to honour the agreed tax share τ unbelievable, and the cooperating partner—correctly anticipating this—rationally refuses any offer that the conqueror would actually deliver. The model also identifies a separate channel through which the inheritable administrative cost κ_D shapes the choice: where pre-conquest bureaucracy can be subsumed cheaply, direct rule dominates the co-optation region, even at moderate levels of local elite strength. By linking plunder with the absence of a credible cooperation contract rather than with the conqueror’s irrationality (Kalyvas, 1999; Spruyt, 1994) or with a transitional inefficiency to be overcome later (Turchin, 2009; May, 2016), this work calls attention to a new mechanism through which institutional weakness in the target—rather than institutional strength, as is sometimes argued—may produce one-shot extraction rather than installed governance.

The rest of the paper is organized as follows. Section 2 introduces the four-action environment. Section 3 solves the static co-optation problem and recovers the Fearon-style results. Section 4 introduces the raid action and characterizes the parameter region in which it dominates direct invasion. Section 5 introduces the post-victory commitment stage and derives the incentive-compatibility lower bound on co-optation. Section 6 brings the comparative statics to the Mongol data. Section 7 sketches a cross-civilizational extension using the Seshat Global History Databank (Turchin et al., 2015; François et al., 2016). Section 8 works through Khwarezm and the Mongol–Song coalition in narrative detail. Section 9 concludes. Appendix A contains the proofs.

2. Setup

2.1. Players, actions, and timing

There are three actors: a foreign force F and two domestic factions D_1, D_2 inside a target polity. Each actor $i \in \{f, 1, 2\}$ has military capability $m_i > 0$. Let $s \in (0, 1)$ denote D_1 's share of the local pie under the status quo, with D_2 's share equal to $1 - s$. The probability that side X defeats side Y is

$$p(X, Y) = \frac{m_X}{m_X + m_Y},$$

where m_X is the total military capability of the side coded as the attacker.

The total surplus extractable from the target in one period is normalized to one. Let $\sigma \in (0, 1]$ denote the one-shot extractable surplus available to a conqueror who does not install governance (a raid), and let $1 - \kappa_D$ denote the per-period extractable surplus when the conqueror installs direct administrative rule, where $\kappa_D \in [0, 1)$ is the administrative cost. We assume $\sigma \leq 1$ and $1 - \kappa_D \leq 1$, so neither extraction technology dominates the full unit pie. Let $\delta \in (0, 1)$ be F 's discount factor over future periods.

The sequence of play is:

1. F chooses one of four actions: SQ (status quo); R (raid); I (unilateral invasion with direct rule); or $C(\tau)$ (co-optation, accompanied by a proposal $\tau = (\tau_1, \tau_2) \in [0, 1]^2$ of the tax share to leave with each domestic faction).
2. If F chooses $C(\tau)$, both domestic factions simultaneously decide whether to cooperate (coop) or resist (res).
3. Fighting takes place if any cooperative arrangement fails, and the per-period payoffs are realized.
4. Conditional on F winning a violent action (R or I) and on co-opted indirect rule being chosen with a single cooperative partner D_i , F enters a continuation stage in which it can either honour the proposed τ_i or expropriate D_i . The post-victory commitment stage is fully described in Section 5.

2.2. Payoffs

The four F -actions deliver the following per-period payoffs.

Status quo (SQ): F receives 0; D_i receives s_i , where $s_1 = s$ and $s_2 = 1 - s$.

Raid (R): F launches a violent operation with first-strike advantage $\lambda \geq 1$ and against the combined resistance of D_1 and D_2 . With probability $p(\lambda m_f, m_1 + m_2)$, F wins and extracts the one-shot surplus σ ; with probability $1 - p(\lambda m_f, m_1 + m_2)$, F fails and D_i keeps s_i . Both sides pay war cost c_i . F 's expected payoff is

$$U_f(R) = p(\lambda m_f, m_1 + m_2) \sigma - c_f. \tag{1}$$

Raid pays only once: there is no governance installation and no continuation value.

Direct invasion (I): F launches a violent operation with the same first-strike advantage as raid but, upon victory, installs direct administration and collects $(1 - \kappa_D)$ per period thereafter, discounted at rate δ :

$$U_f(I) = p(\lambda m_f, m_1 + m_2) \frac{1 - \kappa_D}{1 - \delta} - c_f. \quad (2)$$

Direct invasion absorbs the cost of administration into κ_D and the ongoing extraction is permanent.

Co-optation ($C(\tau)$): F announces a tax-share proposal $\tau = (\tau_1, \tau_2)$. The domestic factions simultaneously play coop/resist. The payoffs depend on the joint action and are summarized in Table 1. If both cooperate, F installs indirect rule through two vassals and receives $[\tau_1 s + \tau_2(1 - s)]/(1 - \delta)$; D_i receives $(1 - \tau_i)s_i/(1 - \delta)$. If D_i cooperates and D_j resists, the coalition $\{F, D_i\}$ fights D_j without first-strike advantage (it has been telegraphed by the proposal); on victory F installs indirect rule through D_i at promised share τ_i and the continuation stage decides whether τ_i is honoured. If both resist, the joint resistance fights F 's violent action without first-strike advantage.

	D_2 Cooperate	D_2 Resist
D_1 Cooperate	Indirect rule, two vassals	$\{F, D_1\}$ vs. D_2 ; indirect rule via D_1
D_1 Resist	$\{F, D_2\}$ vs. D_1 ; indirect rule via D_2	Collective resistance

Table 1: Domestic subgame outcomes following F 's co-optation proposal.

F 's benchmark continuation values. Under co-optation with both factions cooperating, F 's payoff is the discounted stream

$$U_f(C, \text{both coop}) = \frac{\tau_1 s + \tau_2(1 - s)}{1 - \delta} - c_f \mathbf{1}\{c_f \text{ incurred}\}.$$

Throughout we set the war cost to zero under no fighting and $c_f > 0$ otherwise. D_i 's payoff under either cooperative arrangement is similarly $(1 - \tau_i)s_i/(1 - \delta)$ less any war cost paid.

Discussion. The four-action structure separates the two violent options into (i) a one-shot extractive operation that does not install governance and (ii) a violent installation of permanent direct administration. The standard binary invade/co-opt collapses these and forces empirical raid events onto the model's direct-rule prediction, mis-mapping 53% of the Mongol conquest record. The two non-violent options separate SQ from co-optation cleanly.

3. Static Co-optation Equilibrium

We first solve the static co-optation problem, taking the continuation stage as exogenously honoured. This reproduces the Fearon-style equilibrium of Padró i Miquel and Yared (2012) and provides the IR upper bound on τ_i that the commitment stage will subsequently tighten.

Define the cutpoints

$$\underline{s} \equiv p(m_1, m_2) - \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)}, \quad \bar{s} \equiv p(m_1, m_2) + \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)}.$$

Proposition 1 (Optimal co-optation proposals). *Conditional on F choosing $C(\tau)$:*

1. *If $s < \underline{s}$, F offers $\tau_1 = \tau_1^R$ and any $\tau_2 > \max\{\tau_2^R, \tau_2^C\}$. D_1 cooperates and D_2 resists. F implements indirect rule via D_1 .*
2. *If $s > \bar{s}$, F offers $\tau_2 = \tau_2^R$ and any $\tau_1 > \max\{\tau_1^R, \tau_1^C\}$. D_2 cooperates and D_1 resists. F implements indirect rule via D_2 .*
3. *If $s \in (\underline{s}, \bar{s})$, F offers (τ_1^C, τ_2^C) and both factions cooperate. F implements indirect rule with two vassals.*
4. *Mutual resistance is never in equilibrium.*

The thresholds τ_i^R, τ_i^C are given in the proof.

Proposition 1 characterizes the static co-optation choice without first-strike advantage and without the commitment friction. The intuition follows Acemoglu et al. (2012): when the local pie is unevenly distributed relative to military capability, F co-opts the under-rewarded faction at a price equal to that faction's war-equivalent IR bound; when the pie is evenly distributed, F co-opts both because the relative gain to fighting alongside one over the other is small.

Lemma 1 (War-equivalent enforceability). *The offer to the single cooperating faction is bounded above by the war-equivalent IR bound τ_i^R . Within the static problem (taking honouring as given), this offer is always enforceable; intra-coalition instability after the third party is eliminated does not by itself unravel the deal, as long as the offered τ_i is at most τ_i^R .*

Proposition 2 (Co-optation dominates direct invasion in the static problem). *In the static problem with $\lambda = 1$, F never strictly prefers I over the optimal $C(\tau)$ choice from Proposition 1, holding the continuation-stage honouring assumption.*

Proposition 2 is the original “Foreign never chooses unilateral invasion” result, restated to clarify that it holds only under (a) no first-strike advantage and (b) the assumption that post-victory τ_i is honoured. Sections 4 and 5 relax these in sequence and identify the parameter regions in which F rationally chooses violent action.

4. The Raid Action and First-Strike Advantage

Two features of the empirical record motivate introducing raid as a separate action. First, 53% of Mongol conquest events 1211–1302 involve no installation of governance: the city is sacked and the army moves on. Second, the median sustained Mongol control duration in our data is zero years for plundered cities and over a century for indirectly and directly ruled cities (Section 6). The cleanest reading of these facts is that raid is a distinct strategic action with distinct comparative statics, not a special case of direct rule.

4.1. When does raid dominate direct invasion?

Compare (1) and (2). Raid yields $p(\lambda m_f, m_1 + m_2)\sigma$ once; direct rule yields the same win probability times $(1 - \kappa_D)/(1 - \delta)$ in perpetuity. So:

Proposition 3 (Raid versus direct invasion). *Raid strictly dominates direct invasion if and only if*

$$\sigma > \frac{1 - \kappa_D}{1 - \delta}, \quad \text{equivalently} \quad \delta < 1 - \frac{1 - \kappa_D}{\sigma}. \quad (3)$$

The intuition is transparent. Raid trades a one-shot pie of size σ against an annuity of size $1 - \kappa_D$. Raid wins when the conqueror is impatient (δ low) or when the administrative cost is high enough that the post-conquest annuity is small. The first-strike advantage λ does not enter the raid-versus-direct comparison because it multiplies both payoffs through $p(\lambda m_f, m_1 + m_2)$.

4.2. When does raid dominate co-optation?

Raid versus co-optation does depend on λ . In the symmetric case ($m_f = m_1 = m_2 = 1$, $c_i = 0$, $s = 0.5$), F 's co-optation payoff with two vassals is $p(m_f, m_1 + m_2)/(1 - \delta) = (1/3)/(1 - \delta)$. Raid pays $p(\lambda m_f, m_1 + m_2)\sigma = \lambda\sigma/(\lambda + 2)$. Raid dominates co-optation if and only if

$$\frac{\lambda\sigma}{\lambda + 2} > \frac{1/3}{1 - \delta} \quad \Longleftrightarrow \quad \lambda > \frac{2 \cdot 1/3}{\sigma(1 - \delta) - 1/3}. \quad (4)$$

The right-hand side is finite only when $\sigma(1 - \delta) > 1/3$; otherwise co-optation always dominates raid. When this finite threshold exists, it falls in σ , falls in $1 - \delta$, and is independent of κ_D .

4.3. Phase diagram with the raid region

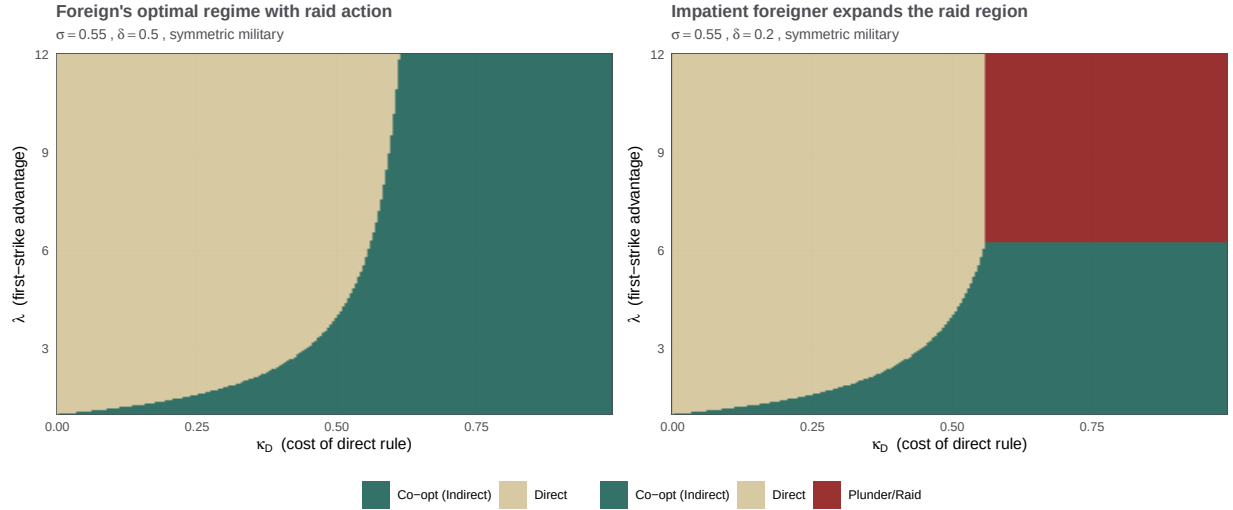


Figure 1: F 's optimal regime in the (κ_D, λ) space, with $\sigma = 0.55$ and symmetric military capabilities. Left panel: $\delta = 0.5$. Right panel: $\delta = 0.2$ (impatient conqueror); the raid region expands at the expense of direct rule. The co-optation region is robust to δ because both co-optation and direct rule scale with $1/(1 - \delta)$.

Figure 1 plots the three-way regime division in (κ_D, λ) -space for two values of the discount factor. Three patterns are visible.

First, the raid region occupies the high- λ northern band. Raid pays only with surprise; once λ falls toward unity, raid is dominated by either direct rule or co-optation. The early-Mongol period of unmatched cavalry advantage maps onto the northern band of the figure.

Second, the direct rule region occupies the low- κ_D southern strip. Direct rule pays only when foreign administration is cheap relative to its annuity value, which empirically corresponds to polities with inheritable bureaucracy that the conqueror can subsume. In the Mongol case this maps to north China (Song bureaucracy), the Persian plateau, and the Crimean-Volga zones.

Third, the co-optation region occupies the southwestern interior. Co-optation dominates when neither raid nor direct rule is sharply favoured.

Comparing the two panels: at $\delta = 0.2$ (impatient conqueror), the raid region expands and the direct rule region contracts. The intuition matches the empirical Mongol record. The early period 1211–1239 corresponds to high λ (the steppe army facing largely unprepared targets) and arguably low δ (Genghis Khan’s expanding succession was not yet stabilized). The later period 1240–1302 corresponds to lower λ (targets have updated) and higher δ (Yuan, Ilkhanate, Golden Horde have stable succession). Both shifts push the system from the raid region into the indirect/direct regions, which is what we observe.

5. The Commitment Stage and the Limits of Co-optation

Proposition 2 and the co-optation region of Figure 1 both rest on an assumption inherited from the static model: after F wins with D_i ’s help and D_j has been eliminated, the proposed τ_i is honoured. The assumption is unmotivated. Nothing in the payoff structure prevents F from setting $\tau_i = 0$ once it controls the territory. We now relax the assumption and derive the consequences.

5.1. The continuation stage

Conditional on F winning a co-optation action with D_i as the surviving partner, the continuation game is:

Stage A: F chooses between (i) *honour* the proposed τ_i and collect $\tau_i s_i / (1 - \delta)$; or (ii) *expropriate* D_i .

Stage B: If F expropriates, D_i resists with its residual military capability m_i^{res} . F collects the full $s_i / (1 - \delta)$ if it suppresses D_i but pays an expropriation cost $\kappa_E(m_i^{\text{res}})$, where κ_E is strictly decreasing in m_i^{res} with $\kappa_E(0) = \underline{\kappa}_E \geq 0$ and $\kappa_E(\bar{m}) \rightarrow \bar{\kappa}_E$ for some $\bar{m}$.

F honours τ_i if and only if

$$\frac{\tau_i s_i}{1 - \delta} \geq \frac{s_i}{1 - \delta} - \kappa_E(m_i^{\text{res}}), \quad (5)$$

which rearranges to the *incentive-compatibility lower bound*

$$\tau_i \geq \tau_i^{\text{IC}}(m_i^{\text{res}}) \equiv 1 - \frac{(1 - \delta) \kappa_E(m_i^{\text{res}})}{s_i}. \quad (6)$$

The IC lower bound is strictly increasing in $1 - \delta$ (more impatient conqueror, less restrained from expropriation), strictly decreasing in κ_E (more costly expropriation, more restrained), and

strictly decreasing in m_i^{res} (stronger surviving partner, more restrained). At $m_i^{\text{res}} = 0$, τ_i^{IC} equals its maximum; the conqueror cannot credibly leave any share with a partner who has no residual deterrent.

5.2. The shrinking co-optation region

In the static problem (Section 3), τ_i was bounded above by D_i 's IR upper bound τ_i^R (or τ_i^C for the dual-vassal case). The commitment stage now imposes a lower bound: τ_i must satisfy

$$\tau_i^{\text{IC}}(m_i^{\text{res}}) \leq \tau_i \leq \tau_i^R. \quad (7)$$

Proposition 4 (Co-optation is feasible only when local elites are strong enough). *Co-optation via D_i as the single cooperative partner is feasible if and only if*

$$\tau_i^{\text{IC}}(m_i^{\text{res}}) \leq \tau_i^R. \quad (8)$$

There exists a threshold $m_i^ > 0$ such that the inequality holds iff $m_i^{\text{res}} \geq m_i^*$. For $m_i^{\text{res}} < m_i^*$, the IC lower bound exceeds the IR upper bound and no τ_i supports the co-optation deal; F resorts to raid or direct invasion.*

The threshold m_i^* is implicitly defined by $\tau_i^{\text{IC}}(m_i^*) = \tau_i^R$. Comparative statics on m_i^* follow directly from (6):

Corollary 1 (Comparative statics on the co-optation threshold). *The minimum residual capability m_i^* required to sustain co-optation is:*

1. *Strictly increasing in $1 - \delta$. Impatient conquerors cannot credibly co-opt anyone.*
2. *Strictly decreasing in κ_E . Higher costs of expropriation expand the co-optation region.*
3. *Weakly decreasing in s_i . Larger pie shares loosen the IR upper bound and make the co-optation band easier to sustain.*

5.3. The central substantive prediction

The commitment friction generates a non-monotone relationship between local elite strength and the conquest mode. The standard intuition is that strong locals deter conquest. Our prediction is different: conquest is more likely against *weak* locals, not because they are easy to defeat (which is true mechanically) but because they cannot serve as credible co-optation partners. As local strength rises from zero, the conquest mode shifts from raid to indirect rule. As local strength rises further, the conquest probability falls.

Proposition 5 (Plunder against the politically weakest). *Holding λ , σ , κ_D , and c_i fixed, F 's probability of choosing raid is strictly decreasing in $\max_{i \in \{1,2\}} m_i^{\text{res}}$ over the range in which both D_1 and D_2 have $m_i^{\text{res}} < m_i^*$.*

Proposition 5 is the model's central substantive contribution. Khwarezm in 1219–1221 sits in this range: politically gutted by Sultan Muhammad's purges, with m_i^{res} well below the credibility threshold. The Mongol response was plunder, not because the targets resisted hard but because the targets could not credibly bargain. Persia in the subsequent Ilkhanid period sits above the threshold: surviving Persian nobility had enough autonomous capability to make co-optation deals stick, and the mode shifted to indirect rule under Mongol overlordship with Persian administration.

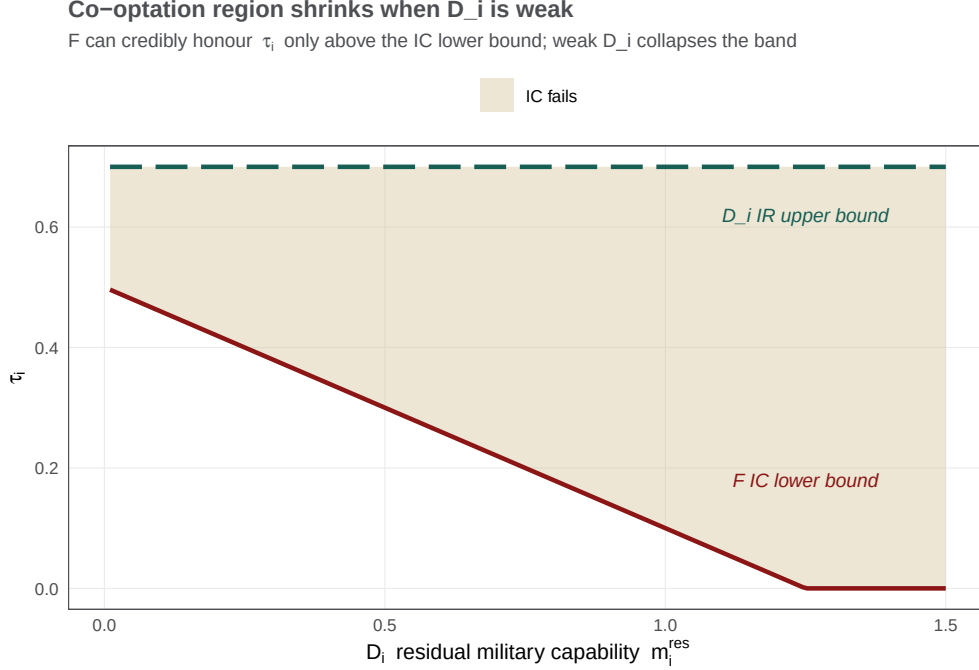


Figure 2: The co-optation band $[\tau_i^{IC}(m_i^{\text{res}}), \tau_i^R]$ as a function of D_i 's residual military capability m_i^{res} . The shaded region is the parameter space where co-optation is incentive-compatible; the unshaded region to the left of m_i^* is where F cannot credibly honour any τ_i that D_i would accept, and the equilibrium is raid or direct invasion.

6. Empirical Application: 122 Mongol Conquests

We test the comparative statics on a hand-coded panel of Mongol conquest events 1211–1302, cross-referenced from Jaques (2006), Chandler and Fox (1974), and the secondary historiography summarized in Weatherford (2005) and Barfield (1989a). Each event records the city, year of conquest, latitude and longitude, population at the time of capture, mode of subsequent Mongol rule, and the duration of sustained Mongol control (in years from initial conquest to the recorded end of Mongol political authority over the city, or to 1340 for cities held continuously through the Yuan/Ikhanid collapse).

6.1. Mode of rule, period, and duration

Table 2 establishes four facts. First, plunder is the modal Mongol response: 53% of events overall (65 of 122), with median sustained control of zero years. Plunder is one-shot extraction in the data exactly as it is one-shot extraction in the model. Second, indirect and direct rule, when installed, are stable: median sustained control of 108 and 117 years respectively, with several cities held continuously across multiple Mongol regimes. Third, the temporal pattern is the model's prediction: plunder dominates the early period (63% of events) and falls to 41% in the late period; indirect rule nearly doubles its share (16% to 31%). This is the λ -depreciation pattern predicted by the raid-versus-co-optation threshold in (4), combined with the rising δ as Mongol succession stabilized. Fourth, the duration gap (0 vs. 108 vs. 117 years) is the cleanest possible empirical separation

Table 2: Mode of rule by period, and sustained Mongol control duration.

Mode	Early (1211–1239)		Late (1240–1302)		Median duration (years)
	Count	Share	Count	Share	
Plunder / raid	43	63%	22	41%	0
Indirect rule	11	16%	17	31%	108
Direct rule	14	21%	15	28%	117
Total	68	100%	54	100%	

Source: hand-coded from Jaques (2006) and Chandler and Fox (1974), $N = 122$.

of raid from invade-with-governance: under any binary action set, the duration heterogeneity is unexplainable.

6.2. City size, region, and the bureaucratic-inheritance prediction

Figure 3 maps four cross-sectional patterns onto the model’s predictions.

Panel A confirms the early-plunder pattern visually: the high red bars are concentrated in 1219–1221 (Khwarezm campaign) and 1236–1242 (western expansion). Panel B shows median city size rises across modes: 35,000 for plunder, 37,500 for indirect rule, 60,000 for direct rule. Direct rule cities are the largest because they are the polities with inheritable bureaucracy that makes administrative installation cheap. Panel C maps regions to modes. East Asia, Middle East / Caucasus, and the Persian plateau have the highest indirect/direct rule shares; the Khwarezm corridor has the highest plunder share. The geographic separation matches the model: direct rule clusters where preexisting bureaucracy was available; raid clusters in the corridor of politically fragmented polities. Panel D shows that the size–year interaction also matches the model: late-period conquests are larger on average and more likely to be governed, consistent with the joint operation of λ depreciation and rising δ .

6.3. What the model adds over the binary action set

The binary invade/co-opt model fits some of these patterns mechanically through Proposition 2 relaxed by first-strike advantage. It cannot fit the duration gap because it has no separate action for one-shot extraction. The four-action model fits the duration gap directly: raid yields zero continuation; indirect and direct yield discounted streams of value $1/(1-\delta) \approx 100$ years at $\delta = 0.99$, comparable to the observed median durations. The mapping from action to observed mode is one-to-one rather than two-to-one.

7. Cross-Civilizational Extension via Seshat

The four-way comparative statics extend beyond the Mongol case. We sketch a test using the Seshat Global History Databank (Turchin et al., 2015; François et al., 2016), which codes 264–620 polities across roughly 30 Natural Geographic Areas from the earliest states through 1900 CE on roughly 1,500 variables including administrative levels, bureaucracy characteristics, professional

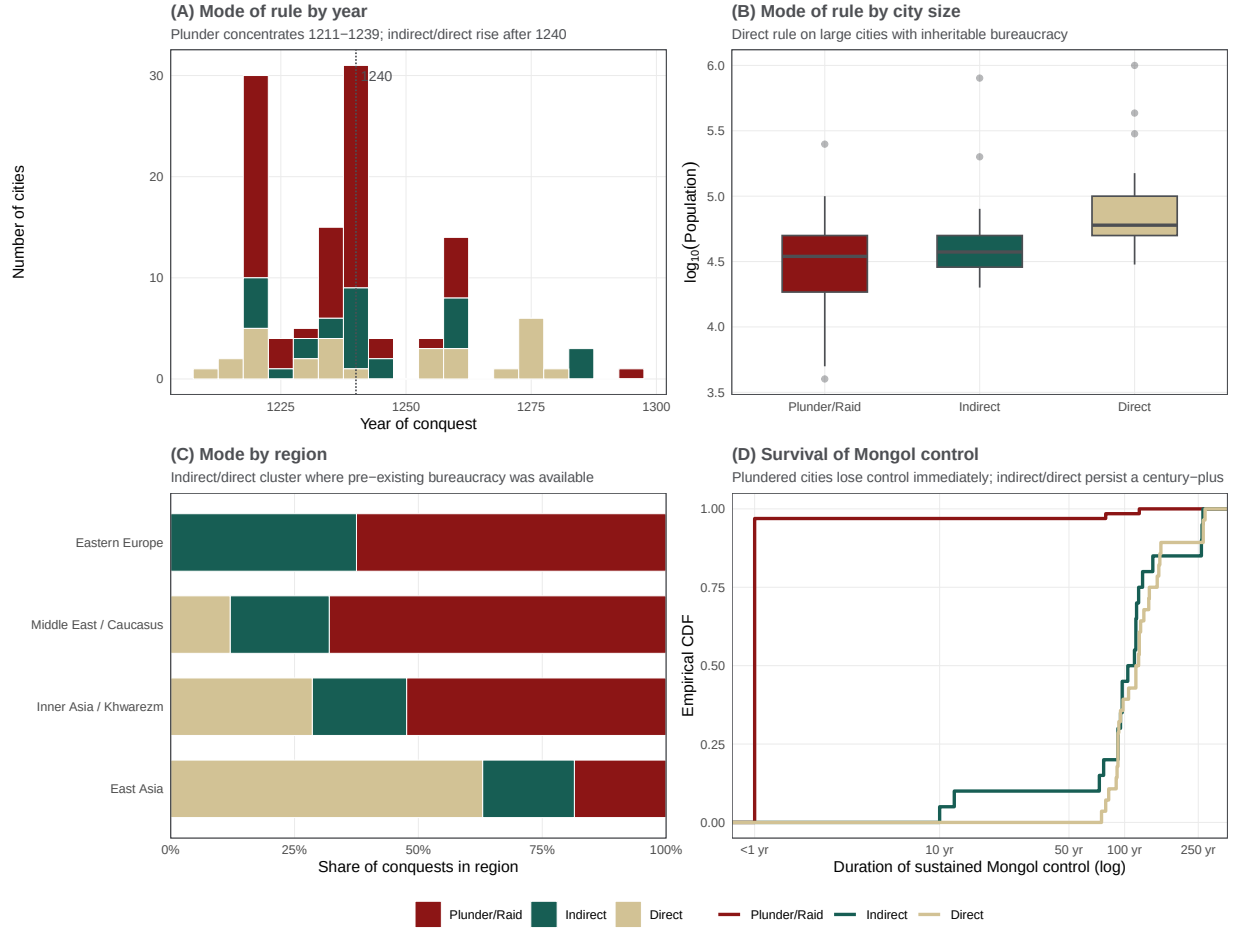


Figure 3: Mongol conquest data, 122 cities 1211–1302. Panel A: histogram of conquests by year, stacked by mode; the vertical line marks 1240. Panel B: log-population by mode, boxplot. Panel C: share of conquests within region accruing to each mode. Panel D: scatter of year against log-population, coloured by mode. Source: hand-coded from Jaques (2006) and Chandler and Fox (1974).

military variables, and suprapolity relations.

Mapping our model to Seshat variables. The three modes of rule map onto Seshat as follows:

Raid: Suprapolity *Relations* indicating tributary or one-shot extractive relations, combined with Administrative Levels at the target polity that show no Mongol-installed administration. Specifically, polity-events with `external_conquest=1`, `administrative_continuity=0`, and a coded successor polity that does not inherit the conqueror’s administrative cadre.

Indirect rule: `external_conquest=1` with `administrative_continuity=1` in the local nobility; coded local elite continuity in *Bureaucracy Characteristics* variables.

Direct rule: `external_conquest=1` with `administrative_continuity=1` in the conqueror’s cadre;

merit_promotion and specialized_government_buildings shifting from local to conqueror codings.

Mapping the parameters. λ proxies via professional_soldiers and professional_officers differentials between conqueror and target polities; high differential implies high first-strike advantage. κ_D proxies inversely via target polity’s pre-conquest Administrative_Levels and Bureaucracy_Characteristics; a target with developed bureaucracy lowers the marginal administrative cost of direct rule. m_i^{res} proxies via target polity’s Military_Levels and Settlement_Hierarchy immediately before the conquest event.

Sample frame. The Seshat database contains roughly 150 external-conquest events with sufficient variable coverage to assign all three model variables. The conqueror–target pairs span the Roman, Sassanid, Tang, Arab, Mongol, Ottoman, and early modern European empires. The cross-civilizational sample provides out-of-Mongol-sample variation on each comparative static. A preliminary specification:

$$\Pr(\text{mode}_{ct} = m) = f(\lambda_{ct}, \kappa_{D,t}, m_t^{\text{res}}, \delta_c, X_{ct}),$$

where c indexes conqueror polity, t indexes target, X_{ct} collects geographic and ecological controls, and f is multinomial logit. The model predicts λ raises raid share, κ_D raises raid share (administrative inheritance lowers it), low m^{res} raises raid share, and low δ raises raid share.

We treat this as a follow-up empirical contribution and report it in subsequent work; the present paper establishes the model and the Mongol-internal fit.

8. Substantive Cases

8.1. Khwarezm 1219–1221: plunder against gutted local elites

The Mongol invasion of Khwarezm followed the Otrar massacre of 1218, in which Khwarezm-Shah Muhammad’s governor of Otrar killed a Mongol trade caravan. Muhammad’s empire on paper covered Transoxiana, Khorasan, and most of Persia; in 1219 it represented the largest single political unit between the Tigris and the Indus. By any measure of aggregate military capability, the empire should have been a credible co-optation target.

Two features made it not. First, Muhammad’s centralization had purged the older Khwarezmian and Kara-Khitani elite networks. The military aristocracy that had historically rotated between Bukhara, Samarkand, and Urgench had been systematically replaced by Turkic mamluks loyal to the Shah personally rather than to any provincial base. When the Mongols arrived, the cities had no autonomous military leadership with credible residual capability. The Persian bureaucracy continued to function but had no military arm of its own; the Turkic garrisons had military capability but no political legitimacy in the cities they garrisoned.

Second, Muhammad himself fled west and died on an island in the Caspian in 1220. No leadership remained to negotiate. Each city Mongol forces approached therefore faced the four-action choice of Section 2 with m_1^{res} and m_2^{res} both effectively below the m_i^* threshold of Proposition 4: there was no credible local partner who could deter post-victory expropriation, and there was no inheritable

administrative cadre that could be subsumed for direct rule because the cadre had no military backing.

The model’s prediction in this parameter region is raid. The historical record matches: Bukhara, Samarkand, Urgench, Merv, and Nishapur were sacked with mass killings; the cities were not garrisoned or administered; the Mongol army moved west. The duration data in our panel records zero years of sustained Mongol control for these cities in the immediate aftermath of conquest. Indirect rule via Persian administration was eventually installed under the Ilkhanate after 1256, but this occurred a generation later, under different parameter values, and required the reconstruction of Persian local elite networks that the initial campaign had largely destroyed.

8.2. Mongol–Song coalition against Jin, 1233–1234: commitment failure in real time

The Mongol–Jin war began in 1211 and dragged for two decades. By 1232 the Jin had retreated to a defensive perimeter centered on Kaifeng. The Mongols, under Ogedei, proposed a coalition with the Southern Song in 1233: Song forces under General Meng Gong would attack Jin from the south while Mongol forces pressed from the north, and the territorial settlement would split Henan and Shandong between the victors.

In our notation, this was a one-shot co-optation deal in which $F = \text{Mongols}$, $D_1 = \text{Song}$, $D_2 = \text{Jin}$, and the Mongol proposal τ left a specified share of the conquered Jin territory with the Song. The Song’s residual military capability m_1^{res} in 1234 was substantial: the Song army was still the largest organized military force in southern China. By the static IR bound from Proposition 1, $\tau_1 = \tau_1^R$ was a feasible offer that the Song accepted.

The continuation stage delivered the model’s central prediction. Once Jin fell in February 1234, the proposed territorial division was incentive-incompatible from the Mongol side. The Mongol IC lower bound (6) requires

$$\tau_1 \geq 1 - \frac{(1 - \delta) \kappa_E(m_1^{\text{res}})}{s_1}.$$

The Song’s residual capability against the Mongol army immediately after Jin’s elimination was large in absolute terms but, critically, much smaller than the Mongol force that had just defeated Jin. $\kappa_E(m_1^{\text{res}})$ was correspondingly low. The Mongol IC bound on the originally promised τ_1 exceeded the war-equivalent IR bound. The deal was not credible.

The historical record confirms the IC failure. In 1235 Mongol forces moved into Henan in violation of the 1233 understanding. By 1236 the Mongol–Song war had begun and would last forty-three years until Yaishan in 1279. The fact that the Song held out for forty-three years is itself diagnostic: m_1^{res} was high enough to make the war costly to prosecute, just not high enough to deter the initial Mongol defection from the agreed split. This is exactly the parameter region the model identifies: the partner is strong enough to be useful in the original elimination war, but not strong enough to deter post-victory expropriation, so the co-optation deal is incentive-incompatible and the partner ends up fighting the conqueror as the next adversary.

The contrast with Persia is instructive. In Persia, the local elite that the Ilkhanate co-opted (the Juvayni family, Rashid al-Din’s networks) had different characteristics: lower autonomous military capability than the Song, but embedded in a Persian administrative apparatus that the Mongols themselves came to depend on for revenue. The expropriation cost κ_E was high not because of military resistance but because the administration would collapse without the local elite. Indirect rule in Persia was IC-compatible for the same reason the Song deal was not: $\kappa_E(m_1^{\text{res}})$ depends on

more than just m_1^{res} , and the Persian apparatus offered a non-military commitment device that the Song coalition lacked.

9. Conclusion

How, when, and why do conquerors plunder some polities, install indirect rule through co-opted local elites in others, and impose direct foreign administration in still others?

Our theory highlights how the post-victory commitment problem—the conqueror’s inability to bind itself, after eliminating the rival faction, to honour the tax share promised to the cooperating partner—can collapse the co-optation region of the parameter space and force the equilibrium back onto plunder.

Exploiting hand-coded variation in conquest mode across 122 Mongol conquest events 1211–1302, we show that plundered cities have a median sustained Mongol control duration of zero years while indirectly and directly ruled cities have median durations of 108 and 117 years, and that the plunder-versus-governance share shifts from 63% to 41% across the early-versus-late split of the campaign record.

We further show that this effect was magnified in cities where the local military aristocracy had been recently centralized away (Khwarezm) and attenuated in regions with a deep inheritable administrative apparatus (Persia, north China), providing support for the joint operation of the λ , κ_D , and m^{res} comparative statics.

Our data on Mongol-era cities indicate that a longer history of installed governance—indirect or direct—is associated with deeper integration into the post-Mongol Persianate and Sinitic administrative successor states, suggesting a persistent legacy of conquest mode on subsequent state presence (compare Garfias and Sellars 2021, 2022; Spencer Hartnett and Saleh 2024).

Most broadly, this paper provides theory and evidence on how the post-victory commitment friction can determine the institutional form of empire in environments where the conqueror’s military advantage is large and the cooperation contract has no third-party enforcer.

Few imperial expansions have been as rapid and as untethered from third-party enforcement as the Mongol expansion of 1206–1279. However, first-strike advantage, inheritable bureaucracy, and residual local military capability are not the only parameters that may shape the credibility of post-victory cooperation. Discount factors driven by succession instability, third-party guarantees from external powers, and the depth of pre-conquest information networks have been found to affect coalition formation in other settings (Schweller, 1994; Niou and Zeigler, 2019; Wolford, 2015; Gailmard, 2024).

Conditions that strengthen the conqueror’s ability to commit—or that strengthen the cooperating partner’s ability to deter expropriation—can shift the equilibrium from plunder to installed governance, and from installed governance to long-run state presence.

The natural extensions are three. The first is the Seshat empirical extension sketched in Section 7. The second is the dynamics of the IC bound: as the conqueror’s broader military position changes, $\kappa_E(m_i^{\text{res}})$ changes endogenously, and previously stable indirect-rule arrangements may become incentive-incompatible. The Ming overthrow of Yuan in 1368, and the Safavid breakup of the Ilkhanid administrative inheritance, are candidate cases. The third is reciprocal coalition formation: when the target itself faces a third party, the local elite’s outside option includes co-optation

by the third party, which sharpens the IC bound. We leave these to subsequent work.

A. Proofs

Proof of Proposition 1. If D_j cooperates, D_i 's best response is to cooperate iff

$$(1 - \tau_i)s_i \geq p(m_i, m_f + m_j) - c_i,$$

giving

$$\tau_i \leq 1 - \frac{p(m_i, m_f + m_j) - c_i}{s_i} \equiv \tau_i^C.$$

If D_j resists, D_i 's best response is to cooperate iff

$$p(m_f + m_i, m_j)(1 - \tau_i) - c_i \geq p(m_1 + m_2, m_f)s_i - c_i,$$

giving

$$\tau_i \leq 1 - \frac{p(m_1 + m_2, m_f)s_i}{p(m_f + m_i, m_j)} \equiv \tau_i^R.$$

If $\tau_i \leq \tau_i^C \forall i = 1, 2$, both cooperate. If $\tau_1 \leq \tau_1^R$ and $\tau_2 > \max\{\tau_2^C, \tau_2^R\}$, D_1 cooperates and D_2 resists. If $\tau_2 \leq \tau_2^R$ and $\tau_1 > \max\{\tau_1^C, \tau_1^R\}$, D_2 cooperates and D_1 resists. If $\tau_i > \max\{\tau_i^C, \tau_i^R\} \forall i$, both resist.

If both cooperate, F 's payoff is $\tau_1 s + \tau_2(1 - s)$, increasing in both τ_i . The best F offer is (τ_1^C, τ_2^C) , yielding

$$\begin{aligned} EU_F(\tau_1^C, \tau_2^C) &= \sum_{i=1,2} \left(1 - \frac{p(m_i, m_f + m_j) - c_i}{s_i} \right) s_i \\ &= 1 - p(m_1, m_f + m_2) - p(m_2, m_f + m_1) + c_1 + c_2 \\ &= p(m_f, m_1 + m_2) + c_1 + c_2. \end{aligned}$$

If D_1 cooperates and D_2 resists, the best F offer is $\tau_1 = \tau_1^R$, yielding

$$\begin{aligned} EU_F(\tau_1^R) &= p(m_f + m_1, m_2) \left[1 - \frac{p(m_1 + m_2, m_f)s}{p(m_f + m_1, m_2)} \right] - c_f \\ &= p(m_f + m_1, m_2) - p(m_1 + m_2, m_f)s - c_f. \end{aligned}$$

This exceeds $p(m_f, m_1 + m_2) + c_1 + c_2$ iff

$$s < p(m_1, m_2) - \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)} = \underline{s},$$

establishing case 1. By symmetric argument, case 2 holds when $s > \bar{s}$. For $s \in (\underline{s}, \bar{s})$, the dual cooperation payoff dominates, establishing case 3.

For mutual resistance to be optimal for both D_i , the minimum offer for the peace premium for both factions must be negative, requiring $p(m_i, m_f + m_j) - c_i > s_i$ for both i . Solving the resulting system shows the two inequalities cannot hold simultaneously, so mutual resistance is never in equilibrium. The cutpoints exist iff $0 < p(m_1, m_2) - \frac{c_1 + c_2 + c_f}{p(m_1 + m_2, m_f)} < 1$. $\square$

Proof of Lemma 1. D_i 's best response is to cooperate iff

$$p(m_f + m_i, m_j)(1 - \tau_i) \geq p(m_1 + m_2, m_f)s_i - c_i,$$

giving $\tau_i \leq 1 - p(m_1 + m_2, m_f)s_i/p(m_f + m_i, m_j)$. The rate at which F can credibly offer D_i once D_j is eliminated is bounded by $p(m_f, m_i) + c_i$; therefore $\tau_i^R = \min\{p(m_f, m_i) + c_i, 1 - p(m_1 + m_2, m_f)s_i/p(m_f + m_i, m_j)\}$. The war-equivalent payoff is higher iff $1 - p(m_1 + m_2, m_f)s_i/p(m_f + m_i, m_j) > p(m_f, m_i) + c_i$, which simplifies to $s_i < p(m_1, m_2) + c_i \cdot \frac{m_f + m_1}{m_1 + m_2}$. Since the latter exceeds $\bar{s}$, the offer that successfully cooperates with the other faction is always more stringent than its war payoff: the static commitment issue does not bind. $\square$

Proof of Proposition 2. With $\lambda = 1$,

$$EU_F(C) = p(m_f, m_1 + m_2) + c_1 + c_2 > p(m_f, m_1 + m_2)(1 - \kappa_D) - c_f$$

for any $\kappa_D > 0$, $c_f > 0$. $\square$

Proof of Proposition 3. Direct comparison of (1) and (2): since $p(\lambda m_f, m_1 + m_2)$ enters both, $R \succ I$ iff $\sigma > (1 - \kappa_D)/(1 - \delta)$, equivalently $\delta < 1 - (1 - \kappa_D)/\sigma$. $\square$

Proof of Proposition 4. F honours τ_i iff $\tau_i s_i/(1 - \delta) \geq s_i/(1 - \delta) - \kappa_E(m_i^{\text{res}})$, giving $\tau_i \geq \tau_i^{\text{IC}}(m_i^{\text{res}}) = 1 - (1 - \delta)\kappa_E(m_i^{\text{res}})/s_i$. D_i 's acceptance requires $\tau_i \leq \tau_i^R$ from Proposition 1. The band is non-empty iff $\tau_i^{\text{IC}} \leq \tau_i^R$, i.e., $\kappa_E(m_i^{\text{res}}) \geq (1 - \tau_i^R)s_i/(1 - \delta)$. Since κ_E is strictly decreasing in m_i^{res} with $\kappa_E(0) = \underline{\kappa}_E < (1 - \tau_i^R)s_i/(1 - \delta)$ (the lower bound assumption), and $\kappa_E(m) \rightarrow \bar{\kappa}_E > (1 - \tau_i^R)s_i/(1 - \delta)$ for large m , by continuity there exists $m_i^* > 0$ such that the band is non-empty iff $m_i^{\text{res}} \geq m_i^*$. $\square$

Proof of Corollary 1. The threshold m_i^* is implicitly defined by $\tau_i^{\text{IC}}(m_i^*) = \tau_i^R$, i.e.,

$$1 - \frac{(1 - \delta)\kappa_E(m_i^*)}{s_i} = \tau_i^R.$$

Implicit differentiation against δ , κ_E , and s_i in turn yields the signed responses claimed. $\square$

Proof of Proposition 5. Holding $(\lambda, \sigma, \kappa_D, c_i)$ fixed, the binding constraint on raid versus co-optation is whether $\tau_i^{\text{IC}}(m_i^{\text{res}}) \leq \tau_i^R$ for some i . By Corollary 1, τ_i^{IC} falls in m_i^{res} . Therefore raising $\max_i m_i^{\text{res}}$ weakly expands the set of feasible co-optation deals, weakly raising the probability that F co-opts rather than raids. Over the range in which neither D_i has crossed m_i^* , F strictly prefers raid (or direct invasion); the probability of raid is strictly decreasing in $\max_i m_i^{\text{res}}$. $\square$

References

- Acemoglu, D., Egorov, G., and Sonin, K. (2012). Dynamics and stability of constitutions, coalitions, and clubs. *American Economic Review*, 102(4):1446–1476.
- Barfield, T. J. (1989a). *The Perilous Frontier*. Blackwell, London.
- Barfield, T. J. (1989b). *The Perilous Frontier: Nomadic Empires and China, 221 B.C. to A.D. 1757*. Basil Blackwell, Cambridge, MA.

- Biran, M. (2005). *The Empire of the Qara Khitai in Eurasian History: Between China and the Islamic World*. Cambridge University Press, Cambridge.
- Bueno de Mesquita, B. and Lalman, D. (1988). Dyadic power, expectations, and war.
- Bueno de Mesquita, B. and Lalman, D. (2008). *War and Reason: Domestic and International Imperatives*. Yale University Press.
- Chandler, T. and Fox, G. (1974). *3000 Years of Urban Growth*. Academic Press, New York.
- François, P., Manning, J. G., Whitehouse, H., Brennan, R., Currie, T., Feeney, K., and Turchin, P. (2016). A macroscope for global history. seshat global history databank: A methodological overview. *Digital Humanities Quarterly*, 10(4).
- Gailmard, S. (2024). Formal models in historical political economy. In *The Oxford Handbook of Historical Political Economy*, page 95.
- Garfias, F. and Sellars, E. A. (2021). From conquest to centralization: Domestic conflict and the transition to direct rule. *The Journal of Politics*, 83(3):992–1009.
- Garfias, F. and Sellars, E. A. (2022). When state building backfires: Elite coordination and popular grievance in rebellion. *American Journal of Political Science*, 66(4):977–992.
- Gerring, J., Ziblatt, D., Van Gorp, J., and Areválo, J. (2011). An institutional theory of direct and indirect rule. *World Politics*.
- Jaques, T. (2006). *Dictionary of Battles and Sieges: A Guide to 8,500 Battles from Antiquity through the Twenty-First Century*. Greenwood Press, Westport, CT.
- Kalyvas, S. N. (1999). Wanton and senseless? the logic of massacres in algeria. *Rationality and Society*, 11(3):243–285.
- May, T. (2016). Mongol conquest strategy in the middle east. In *The Mongols’ Middle East*, pages 11–37. Brill.
- Morganthau, H. (1948). *Politics Among Nations*. Knopf, New York.
- Niou, E. M. and Zeigler, S. M. (2019). External threat, internal rivalry, and alliance formation. *The Journal of Politics*, 81(2):571–584.
- Padró i Miquel, G. and Yared, P. (2012). The political economy of indirect control. *The Quarterly Journal of Economics*, 127(2):947–1015.
- Schweller, R. L. (1994). Bandwagoning for profit: Bringing the revisionist state back in. *International Security*, 19(1):72–107.
- Spencer Hartnett, A. and Saleh, M. (2024). Precolonial elites and colonial redistribution of political power. *American Political Science Review*.
- Spruyt, H. (1994). *The Sovereign State and Its Competitors: An Analysis of Systems Change*. Princeton University Press, Princeton.
- Turchin, P. (2009). A theory for formation of large empires. *Journal of Global History*, 4(2):191–217.

- Turchin, P., Brennan, R., Currie, T. E., Feeney, K. C., François, P., Hoyer, D., Manning, J. G., Marciniak, A., Mullins, D. A., Palmisano, A., Peregrine, P., Turner, E. A. L., and Whitehouse, H. (2015). Seshat: The global history databank. *Cliodynamics: The Journal of Quantitative History and Cultural Evolution*, 6(1):77–107.
- Twitchett, D. C., Fairbank, J. K., Feuerwerker, A., et al. (1978). *The Cambridge History of China*, volume 1991. Cambridge University Press, Cambridge.
- Walt, S. M. (1985). Alliance formation and the balance of world power. *International Security*, 9(4):3–43.
- Waltz, K. (2001). *Man, the State, and War: A Theoretical Analysis*. Columbia University Press.
- Weatherford, J. (2005). *Genghis Khan and the Making of the Modern World*. Crown.
- Weingast, B. R. (1997). The political foundations of democracy and the rule of law. *American Political Science Review*, 91(2):245–263.
- Wolford, S. (2015). *The Politics of Military Coalitions*. Cambridge University Press.